

Transformers is a powerful way of stacking neural layers that allows dynamic communication between sequence elements

They process all sequences of data in parallel using self-attention mechanisms rather than step-by-step recurrence

Self-attention – Query, Key Value → works in parallel matrix operation

Query means → what am I looking for

Key → Here is what I have (Information provide are stored here)

Value → Content to share (carries the meaning of words)

Each token communicates each other tokens and decides which one are important and capture context efficiently

Encoder and decoder

Attention layer (where all the tokens interact) and Feed forward/MLP layer (each token privately refines its representation)

Input flow in transformer → Tokenizer splits the texts into smaller units called tokens. Then the input tokens are embedded, i.e. transformed into numerical vectors intended to capture their semantic meaning after training.

Transformer has no sense of order by default. So, we add positional information to embeddings to introduce a sense of order among tokens like special patterns added to embeddings to tell the where each token is in sequence

At each step attention mixes information across tokens and Feed forward layer filters them separately

After normalizing with softmax function and turn them into attention weights, these weights like focus levels

Tokens receive high attention based on the dependency and relativity, and few tokens get low attention

Overall, each token gathers info by taking weighted sum of all the values vectors where the weight comes from those attention scores to make richer context meaning.